

Crayola Commotion Instructions

Setup Instructions:

The Flip Chart:

- There is a 3M Post-It Easel Flip Chart (or something like that) in my office. This was purchased for use at our first Retreat after Marie was hired. And we used something like 1 page, so there should be plenty remaining. Since it is long and flat, I think it is slid behind a file cabinet in the office, partly protruding. The idea is that the full images are made to fit this size, so it can stand on an easel (perhaps needing to be taped steady?) and use the front page for an image. After that, flip to the next page, or tear off the completed image. Also, think about giving the image away at the end, unless you have a good use for it – and you might!
- The flip chart in the picture above is 25" wide by 30" high, and *I hope* that is the size of the flip chart in my office. All calculations ahead are based on that assumption.



The Mini-Pages:

- Before drawing the grid on the flip chart, it is important to make sure the mini-pages are appropriately sized. The Word mail-merge template has cells in the grids that are intended to be squares with sides measuring ~0.75"-0.80". This will allow the full image to fit on one page. (If the cells are 0.75", the picture will be 24" x 27". If they are 0.8", it will be 25.6" x 28.8". That is a bit wide for a page 25" wide, but shouldn't be a problem to stick over the right edge a bit. If it gets wider, though, the chart won't be able to support the rightmost column.
- Note that printing the mail-merged mini-pages may produce cells larger or smaller than expected, so it may be necessary to use the "scale" feature of the print menu to produce cells appropriately sized.
- Now the fun part: cutting out the mini-pages. There are 6 grids per page, and lots of margin space to the right and bottom. Leave the margins there – that is, don't cut out the 4x3 grid exactly. Instead, cut out along the top and left edge, but leave the space to the right and the bottom. These will prove useful when the colored mini-pages are attached.

The lower margin is particularly important, because it contains the page number (e.g., A-1).

The Grid:

- Once you have the mini-pages printed at an appropriate size, create a grid on the flip chart, using the actual width of the printed grids, e.g., 3" if the cells are 0.75" each. The mini-pages contain 4x3 grids. There are 8 columns (A-H) and 12 rows (1-12), so there are 96 mini-pages, labeled A-1, A-2, ... H-11, H-12. The full image is 32 pixels wide by 36 pixels high. The Flip Chart needs to be labeled accordingly:

A-1	B-1	...	H-1
A-2	B-2	...	H-2
...	...	...	...
A-11	B-11	...	
A-12	B-12	...	H-12

- A yardstick should be suitable for this task – thank your lucky stars you aren't trying to draw a 13' x 7' grid on a whiteboard.
- You can draw the gridlines and write the labels dark enough to be viewed easily, because they will eventually be covered by colored mini-pages that will obscure them.

Color Guide:

The participants need to know which colors the numbers stand for. I created two files that can be printed, named "COLOR_TABLE_filename_wNAMES.html", which show the number, corresponding color, and the *name* of the corresponding color, which will be very important because some colors are a bit too similar, and there's always the possibility of color-blind participants.

Crayon Prep:

I like to gather all my colors together and put them in separate baggies with numbers on them before the event. In this case, since you have two different images, and (unfortunately) different color tables, which means (for instance) Red is #4 for one exercise and #0 for the next. If there is a contention problem like this, you might be able to substitute a similar color, e.g., use Scarlet in one image and Red in the other – but you'll need to change the names in the COLOR_TABLE

file. Sorry! The numbering comes straight from the Python Pillow library, and I didn't think of this before, because I've never done two different images on the same day. Ugh.

“Cheat Sheets”:

When running the activity, it helps to have a master sheet showing what is supposed to be happening on the easel. So:

- For each image, there is web page called “ColorMap_*filename*.html” for each image, that shows what the final image is supposed to look like, and also showing the colors assigned to the numbers. Print this page for the people running the activity.
- For each image, there is a data sheet, “Post-It_*filename*_datasheet.xlsx” showing the numbers for each page. This is really helpful in case a mini-page is colored incorrectly and needs to be replaced. Just copy the information onto a blank mini-page. The first two columns are the Column and Row (e.g., A-3) and the rest just need to be copied *across each row*, **not** down each column, to create a fresh mini-page. Print this page for the people running the activity.

Instructions for People Running the Activity:

The following items are needed:

- Easel
- 3M Flip Chart with Grids drawn on it, labeled A-1, A-2, ... H-11, H-12.
- For each image to be assembled:
 - o Packet of 96 mini-pages, featuring grids with numbers on them, and “This is Page *m-n*” in the bottom-right corner., describing where on the Flip Chart the completed mini-page should be attached.
 - o Containers of crayons, labeled with numbers 0..N.
 - o COLOR TABLE, to be posted prominently, so students know what colors are assigned to the numbers 0..N. In addition to the number, the table also shows the name of the color, e.g., “Carnation Pink”.
 - o A “cheat sheet”: a similar page, *not* to be posted prominently, showing what the final image is supposed to look like, and also showing the colors assigned to the numbers. You can use this as your own guide to make sure the image looks like it's supposed to. It usually isn't too hard to tell if one of the pages has colors in the wrong places. But don't show the students the final image, because it helps to have an air of suspense while the image is being assembled.
 - *Note:* there are some occasional unexpectedly different colored pixels in the images where there's something that's a solid color, (a) just to add a

little texture to the image, and (b) to reduce the number of mini-pages that are all the same color. There are probably still a few single color mini-pages, though.

- o Printout of a “data sheet” spreadsheet showing the numbers for each page.
- o Packet of blank mini-pages, which can be used if someone messes up their original mini-page and needs to do a new one. In this case, use the spreadsheet printout: Just copy the information onto a blank mini-page. The first two columns are the Column and Row (e.g., A-3) and the rest just need to be copied *across each row*, **not** down each column, to create a fresh mini-page. I’d suggest printing this as a spreadsheet, with appropriate column labels.

Below are instructions for you to give to participants.

Instructions for the Participants:

Explain to students that all information contained on a computer, including documents, images, music, etc., is contained as numbers. *No need to talk about binary numbers here.* And computer programs, video games, web browsers, and other applications read the numbers and process them as appropriate, in order to produce what we see on our screens.

While there are many ways to store an image on a computer, one way is with the GIF format, which maintains a COLOR TABLE of a collection of colors used in the image, and assigns a number to each color. This way, it is very simple to represent an image as a table of numbers.

Today, we are re-creating a GIF image, using the numbers a computer uses to store it. The image has been subdivided into mini-pages of 4x3 grids, with numbers in it. Use the COLOR TABLE to find the colored crayons and fill in the squares accordingly.

Attaching the mini-pages to the easel:

- Once the mini-page is colored, it can be attached to the easel sheet. The “colored-in” part of the mini-page is the visible portion. The “margin” part should be covered by the next mini-page, across or down. That is, when A-2 is added to the easel, it should cover the margin part at the bottom of A-1. (That means when A-1 is added to the easel, its margins will extend over the grid lines to the right (for B-1) and below (for A-2)).
- The mini-pages to the right and bottom can be taped so that they overlap these margins, which also provide a precise guide for the next page.

In case of errors:

Use the “cheat sheet” and the spreadsheet to produce a replacement copy of a mini-page that needs to be redone. You can put a replacement sheet directly on top of the sheet it’s replacing, rather than try to remove the old one. You will need to fold under or tear off the margins on replacement sheets.